

BRADLEY D. SCHMIDT

SOFTWARE ENGINEERING STUDENT

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SUMMARY

Enthusiastic Software Engineering student with a strong foundation in programming, particularly in C++, Python, and JavaScript. Experienced with microcontrollers, Linux, and Git. Looking to apply problem solving skills to make a meaningful difference.

EDUCATION

Thompson Rivers University

Kamloops, BC, Canada

3rd Year Bachelor of Engineering in Software Engineering (GPA: 3.44)

SKILLS

Technical Skills

Programming Languages: C++, Python, JavaScript, SQL, VHDL

Developer Tools: Linux, Git, VS Code, Vim, Android Studio, Oracle SQL Developer

AI and ML: Google Gemini AI, OpenAI API

Frameworks: React, React Native, Node.js

Platforms: Firebase, GitHub

Hardware: Arduino, Raspberry Pi, Esp8266, FPGA

Personal Skills

Innovative Problem Solving: Great at identifying unconventional solutions to complex challenges

Analytical Thinking: Strong logical reasoning and ability to break down complex problems systematically

Intellectual Curiosity: Rapid learner who thrives on exploring abstract concepts

Independent: Self-driven and good at finding creative solutions without relying on current methods

PROJECTS

AquaFlora Mobile App | React Native, Google Gemini AI, Python, Realm Database

Jul. 2023 – Present

- Developed a Mobile app for aquarium management, for users to track water parameters, reminders, and fish stocking.
- Communicated with interested people in order to discover desired features.
- Integrated Google Gemini AI for image recognition (fish, plants, disease detection) and developed and fine tuned a chatbot for aquarium care guidance using Gemini API.
- Created a Python based scraper to pull and verify a database of over 650 freshwater fish species for the app's knowledge base.
- Implemented monthly subscriptions and integrated Google Calendar for intuitive reminders

FPGA Calculator | VHDL, Vivado, ModelSim, Basys 3 FPGA

Nov. 2024 – Nov. 2024

- Used Basys 3 FPGA to make a four function calculator.
- Interfaced the FPGA with 4-7 segment display, and PMOD KYPD keypad
- Implemented addition, subtraction, multiplication and division using behavioural modeling in VHDL.
- Completed this with a partner as part of the CENG 3010 Digital Systems Desgn class at TRU.

Gardening Business Website | React, HTML, CSS

Aug. 2023 – Sep. 2023

- Worked with business owner to come up with website requirements.
- Worked through design and creation with owner.
- Build a strong foundation of HTML, CSS, and React before working on the AquaFlora mobile app.

URRepair | C++, React, JavaScript

Oct. 2024 – Present

- Developed a web based platform connecting contractors to job opportunities as part of a group project.
- Created backend (C++), and frontend interface (Node.js)
- Worked in a group of three to implement three sorting algorithms, and three data structures to ensure optimal function.

VR Gloves | ESP8266, Potentiometers, Servo Motors

Jun. 2024

- Assembled a force feedback, hand-tracking glove for VR using open source designs.
- Utilized ESP8266 for wi-fi connectivity and tested with VR integration, gaining experience in hardware to software interaction.

WORK EXPERIENCE

Maintenance

Service Canada

Mar. 2020 – Present

Kamloops, BC, Canada

- Maintained building cleanliness during peak COVID lockdown, demonstrating responsibility and attention to detail.
- Performed tasks such as carpet shampooing, pressure washing, and general upkeep to ensure a safe environment.

Concession Worker

Senor Froggy

Feb. 2023 – Feb. 2024

Kamloops, BC, Canada

- Supported the main cook in a high paced concession environment during peak hockey event hours.
- Worked efficiently as part of a tight knit team under pressure, enhancing teamwork and time management skills.